

IRONHACK

Final Capstone Project

Data Engineering Track

Duration: 40 Hours

Level: Intermediate to Advanced Data Engineering

Phase 1	SQL & Fraud Analytics — Snowflake (Bank Fraud Dataset)
Phase 2	Advanced Pandas — Telecom Customer Churn Dataset
Phase 3	Airflow & dbt — Pipeline Orchestration
Phase 4	NiFi, Kafka, Spark — Real-Time Streaming & Processing
Phase 5	End-to-End Capstone — Real-Time Social Media Analytics Platform

Phase 1 — SQL & Fraud Analytics

Bank Fraud Dataset | Snowflake SQL

Module 1: Data Exploration & Profiling

Q1. Calculate the following aggregate metrics:

- Total Transactions
- Total Customers
- Total Fraud Transactions
- Fraud Percentage

Q2. Find the top 10 countries by transaction volume.

Q3. Find the top 10 cities generating the highest transaction value.

Q4. Generate a complete data profiling report showing:

Column Name	Distinct Values	Null Count
<i>(to be completed)</i>		

Q5. Determine the percentage distribution of transactions by:

- Payment Method
- Device Type
- Merchant Category

Module 2: Customer Behavior Analytics

Q6. Identify the top 20 customers by SUM(transaction_amount).

Q7. Calculate average transaction amount by age group:

- 18–25
- 26–35
- 36–50
- 51–65
- 65+

Q8. Determine which age group has the highest fraud rate.

Q9. Find customers with:

- High Balance
- Low Transaction Frequency
- Potential dormant accounts

Q10. Calculate average account balance and credit score by country.

Module 3: Fraud Pattern Analysis

Q11. Calculate fraud rate by Merchant Category.

Q12. Determine fraud rate by Payment Method.

Q13. Determine fraud rate by Device Type.

Q14. Find the most common fraud type. Output columns:

- fraud_type
- count
- percentage

Q15. Analyze fraud transactions occurring during:

- Night vs. Day
- Weekend vs. Weekday

Module 4: Advanced Fraud Investigation

Q16. Find transactions where transaction_amount exceeds:

```
AVG(transaction_amount) + 3 * STDDEV(transaction_amount)
```

Identify and report potential anomalies.

Q17. Identify customers having more than 3 failed attempts AND Fraud = 1.

Q18. Determine whether recent PIN changes increase fraud probability. Compare:

- pin_changed_recently = 1
- pin_changed_recently = 0

Q19. Analyze fraud rates for:

- International Transactions
- Domestic Transactions

Q20. Find customers who performed all of the following, then rank by risk:

- International Transaction
- Night Transaction
- Failed Attempts > 2

Module 5: Window Functions

Q21. Rank customers by transaction value within each country.

Use: RANK()

Q22. Find the top 5 highest-value transactions in every merchant category.

Use: ROW_NUMBER()

Q23. Calculate cumulative transaction amount per customer over time.

Use: SUM() OVER()

Q24. Create customer transaction deciles using NTILE(10). Classify customers into:

- Top 10%
- Top 20%

... based on total transaction value.

Module 6: Capstone Fraud Analytics Challenge

Q25. Build a single executive fraud report. Use CTEs, CASE Statements, Window Functions, and Aggregations in one Snowflake solution.

Transaction KPIs	Fraud KPIs	Risk KPIs
• Total Transactions • Total Transaction Value • Average Transaction Amount	• Total Fraud Cases • Fraud % • Top Fraud Type	• Highest Risk Country • Highest Risk Merchant Category • Highest Risk Payment Method

Customer KPIs

- Average Credit Score of Fraudsters
- Average Account Balance of Fraudsters
- Average Failed Attempts Before Fraud

Q26. Create a View: VW_HIGH_RISK_TRANSACTIONS

Include records where:

- Transaction is International
- Transaction occurred at Night
- Failed Attempts > 2

Q27. Create a Materialized View for daily fraud reporting.

Q28. Detect "Suspicious Customers" using only SQL. Criteria:

- Above-average transaction amount
- International transaction
- More than 2 failed attempts
- Credit score below 600

Phase 2 — Advanced Pandas

Telecom Customer Churn Dataset

Solve the following advanced-level Pandas and Data Analytics exercises using the Telecom Customer Churn dataset.

Module 1: Data Profiling & Quality

Q1. Create a complete Data Quality Report showing: data type, null count, null percentage, distinct count, duplicate count, and min/max values where applicable.

Q2. The TotalCharges column is stored as an object. Identify invalid values, convert it to numeric, and quantify affected records.

Q3. Create a function that automatically categorizes columns into: Numerical, Categorical, Binary, and Identifier groups.

Q4. Identify customers where TotalCharges differs from MonthlyCharges × Tenure by more than 10%. Investigate possible causes.

Module 2: Advanced Customer Analytics

Q6. Calculate churn rate by Gender, Senior Citizen, Partner, and Dependents. Rank the highest-risk customer groups.

Q7. Create customer segments based on tenure and analyze churn across segments.

Q8. Identify the top 10 customer profiles most likely to churn using combinations of Contract, Payment Method, and Internet Service.

Q9. Calculate Customer Lifetime Value (CLV = MonthlyCharges × Tenure) and compare churned vs. retained customers.

Q10. Create a Customer Risk Score using Contract Type, Tenure, Tech Support, and Online Security.

Module 3: Revenue Analytics

Q11. Calculate annualized revenue loss due to churn.

Q12. Perform Pareto analysis to identify the top 20% of customers contributing the highest revenue. (OPTIONAL)

Q13. Calculate ARPU (Average Revenue Per User) by Contract Type, Internet Service, and Payment Method.

Q14. Create a revenue leakage report showing high Monthly Charges, low Tenure, and Churn = Yes.

Q15. Find customers paying more than 1.5 standard deviations above the mean monthly charge.

Module 4: Service Usage Intelligence

Q16. Determine which service combination has the lowest churn.

Q17. Identify services most strongly associated with churn reduction.

Q18. Build a Service Adoption Score and analyze its relationship with churn.

Q19. Find customers with Fiber Optic internet, no Tech Support, and no Online Security. Calculate their churn rate.

Q20. Determine the most profitable service bundle.

Module 5: Advanced Pandas Challenges

Q21. Create a churn prediction feature dataset using Pandas feature engineering.

Q23. Create a retention dashboard dataset with retention KPIs. (OPTIONAL)

Q24. Build a rule-based churn detection engine with Low, Medium, High, and Critical risk levels.

Phase 3 — Airflow & dbt

Pipeline Orchestration & Data Transformation

Create Airflow DAGs and dbt models for each of the following exercises.

Q1 — First Airflow Pipeline

Objective: Create a DAG that runs daily at 8:00 AM

Tasks:

1. Create a DAG named first_pipeline.
2. Add a task that prints 'Data Pipeline Started'.
3. Add a task that waits for 10 seconds.
4. Add a final task that prints 'Data Pipeline Completed'.
5. Validate DAG execution from the Airflow UI.

Q2 — Dynamic Task Mapping

Objective: Generate tasks dynamically

Tasks:

1. Create a list of countries.
2. Use Dynamic Task Mapping.
3. Process each country independently.
4. Verify parallel execution.

Expected Outcome: Separate tasks are generated automatically.

Q3 — Branching Workflow

Objective: Implement conditional execution

Tasks:

1. Create a BranchPythonOperator.
2. If today is Monday, run Full Load.
3. Otherwise, run Incremental Load.
4. Merge both branches into a final task.

Expected Outcome: Correct branch executes based on the date.

Q4 — Email Notification

Configure a DAG that sends an email when a task fails.

Q5 — Snowflake ETL Pipeline

Objective: Load and transform data in Snowflake

Tasks:

1. Upload a CSV file to Snowflake.
2. Create a SnowflakeOperator task.
3. Execute transformation SQL.
4. Generate a row-count report.

Expected Outcome: Transformed data available in Snowflake.

Q6 — API Data Ingestion

Objective: Consume REST APIs with Airflow

Tasks:

1. Call a public API.
2. Store JSON output.
3. Parse the response.
4. Save data into a database table.

Expected Outcome: API data successfully loaded.

Q7 — Customer Churn Data Mart

Objective: Build an end-to-end dbt project

Tasks:

- Create staging models.
- Create intermediate models.
- Create fact and dimension tables.
- Generate customer churn KPIs.
- Document models.

Expected Outcome: Star schema and churn mart available.

Q8 — Airflow + dbt Integration

Objective: Orchestrate dbt using Airflow

Tasks:

1. Create Airflow DAG.
2. Execute dbt run.
3. Execute dbt test.
4. Capture logs.
5. Configure failure notifications.

Expected Outcome: Fully orchestrated ELT workflow.

Q9 — Banking Fraud Detection Project

Objective: Build a fraud analytics pipeline

Tasks:

- Load transaction data.
- Apply fraud detection rules.
- Flag suspicious transactions.
- Generate fraud summary tables.
- Schedule daily execution.

Expected Outcome: Fraud alerts generated automatically.

Q10 — Enterprise-Grade ELT Pipeline

Objective: Create an enterprise-grade ELT pipeline

Tasks:

1. Ingest CSV files.
2. Load into Snowflake.
3. Build dbt staging layer.

4. Build marts.
5. Run tests.
6. Schedule with Airflow.
7. Generate business reports.

Expected Outcome: Production-style ELT pipeline.

Phase 4 — NiFi, Kafka & Spark

Real-Time Streaming & Data Processing

- Q1.** What is a FlowFile in Apache NiFi? Describe its two main components and explain what happens to a FlowFile when a processor fails.
- Q2.** Build a NiFi flow that reads CSV files from a local folder, splits each row into individual FlowFiles, and writes them to a destination folder with the filename format: row_<n>.txt.
- Q3.** Explain the difference between 'Back Pressure Object Threshold' and 'Back Pressure Data Size Threshold' in a NiFi connection. When would you tune each?
- Q4.** Design a NiFi dataflow that polls a REST API every 5 minutes, extracts a JSON array from the response, routes records where 'status' is 'ERROR' to an alert folder, and the rest to a success folder.
- Q5.** You need to ingest data from a Kafka topic in NiFi and merge every 1,000 records (or every 10 seconds, whichever comes first) into a single JSON array before writing to HDFS. How do you build this flow and what are the failure handling strategies?
- Q6.** Explain the role of a Kafka broker, a topic, a partition, and a consumer group. Draw or describe how they relate to each other.
- Q7.** A Kafka consumer group has 5 consumers subscribed to a topic with 3 partitions. What happens? What if you later add 2 more partitions — describe the rebalance process.
- Q8.** Write a Python Kafka producer that reads rows from a CSV file and sends each row as a JSON message to a topic. Implement error handling and ensure at-least-once delivery using `acks='all'`.
- Q9.** Design a Kafka Streams application that reads from a 'transactions' topic, filters transactions above \$10,000, groups by customer_id, and computes a running total per customer, writing results to a 'high-value-customers' topic.
- Q10.** What is the difference between a Spark transformation and an action? Give 2 examples of each and explain why Spark uses lazy evaluation.
- Q11.** You have two DataFrames: 'orders' (order_id, customer_id, amount) and 'customers' (customer_id, name, city). Write a Spark job that joins them, finds the top 5 customers by total spend per city, and writes the result as Parquet partitioned by city.
- Q12.** Write a Spark job that reads a large JSON dataset, applies a custom UDF that classifies each record into 'high', 'medium', or 'low' risk using a rule set, then writes each class to a separate Snowflake table using the Spark-Snowflake connector.

Q13. What is the difference between Spark Streaming (DStreams) and Structured Streaming?

Q14. Write a Structured Streaming job that reads from a Kafka topic 'page-views', parses JSON events with fields (user_id, page, timestamp), and computes the count of page views per page in a 5-minute tumbling window. Write results to console.

Q15. A Structured Streaming job using a 10-minute window is missing events that arrive 3 minutes late. How do you handle late data in Spark Structured Streaming? What is watermarking?

Q16. Design a Structured Streaming pipeline that reads IoT sensor data from Kafka (sensor_id, temperature, ts), detects anomalies (temperature > 3 standard deviations from the 1-hour rolling mean per sensor), and writes alerts to a Kafka alerts topic AND to Snowflake for long-term storage.

Q17. End-to-end mini project: Use NiFi to ingest CSV sales files from a folder and publish each row as a Kafka message. Write a Spark Streaming job to consume from Kafka, aggregate sales by product every 2 minutes, and write the result to Snowflake. Build a Power BI dashboard on top of that Snowflake table.

Q18. A retail company wants to build a real-time analytics platform. Data arrives via point-of-sale systems. Describe which tools you'd use at each stage (ingestion, streaming, storage, visualization) and justify your choices.

Q19. Design and implement a late-data handling strategy: IoT sensor events flow through Kafka into Spark Structured Streaming. Some events arrive up to 10 minutes late. Aggregated results must land in Snowflake. Power BI must show both the latest aggregates AND a 'corrections history' when late data updates a previous window. Describe the architecture and key implementation decisions.

Phase 5 — End-to-End Capstone Project

Real-Time Social Media Analytics Platform

Business Scenario

Build a real-time social media analytics platform similar to Instagram, TikTok, LinkedIn, or X. Users generate events such as posts, likes, comments, shares, follows, video views, and profile visits. The platform must ingest, process, enrich, store, and analyze these events in real time.

Technology Stack

Python Event Simulator	
Kafka	
PySpark Streaming	
Snowflake	
Power BI	
Docker	
Terraform (optional)	

Architecture

Simulator → Kafka → PySpark Streaming → Snowflake RAW → CURATED → ANALYTICS → Dashboard (Power BI / Looker)

Step 1 – Project Planning & Architecture

Tasks:

- Draw architecture diagram.
- Identify event types: POST_CREATED, LIKE, COMMENT, SHARE, FOLLOW, VIDEO_VIEW, PROFILE_VISIT.
- Design RAW, CURATED and ANALYTICS layers in Snowflake.
- Define business KPIs and success criteria.

Deliverables:

- Architecture diagram
- Data model

- Project plan

Step 2 – Build Social Media Event Simulator

Tasks:

- Generate at least 100 users and 500 posts.
- Generate realistic events continuously.
- Publish at least 1,000 events per minute.
- Create viral-event simulation capability.

Deliverables:

- Python simulator
- Sample event payloads

Step 3 – Kafka Ingestion Layer

Tasks:

- Create Kafka topics.
- Publish simulator events to Kafka.
- Validate ingestion using Kafka consumers.
- Document topic design.

Deliverables:

- Working Kafka pipeline

Step 4 – Real-Time Processing with PySpark

Tasks:

- Consume Kafka streams.
- Parse JSON payloads.
- Perform data quality checks.
- Handle malformed records.
- Create streaming transformations.

Deliverables:

- Clean streaming dataset

Step 5 – Snowflake RAW Layer

Tasks:

- Create SOCIAL_MEDIA_DB.
- Create RAW schema and RAW_EVENTS table.
- Load streaming data into Snowflake.

Deliverables:

- RAW layer populated

Step 6 – Data Enrichment**Tasks:**

- Create USER_DIM and POST_DIM tables.
- Join streaming events with dimensions.
- Create CURATED_EVENTS dataset.

Deliverables:

- Enriched curated dataset

Step 7 – Trending Hashtag Analytics**Tasks:**

- Calculate top hashtags for 1, 5 and 15-minute windows.
- Store results in analytics tables.

Deliverables:

- TRENDING_HASHTAGS table

Step 8 – Viral Content Detection**Tasks:**

- Identify posts receiving more than 500 likes within 5 minutes.
- Generate alerts and analytics outputs.

Deliverables:

- VIRAL_POSTS table

Step 9 – Influencer Ranking**Tasks:**

- Create engagement scoring model.
- Rank creators based on likes, comments and shares.

Deliverables:

- INFLUENCER_RANKING table

Step 10 – Sentiment Analytics

Tasks:

- Classify comments into Positive, Neutral and Negative.
- Generate sentiment trends.

Deliverables:

- COMMENT_SENTIMENT analytics

Step 11 – Dashboard Development

Tasks:

- Build Engagement Dashboard.
- Build Viral Content Dashboard.
- Build Influencer Dashboard.

Deliverables:

- Dashboards